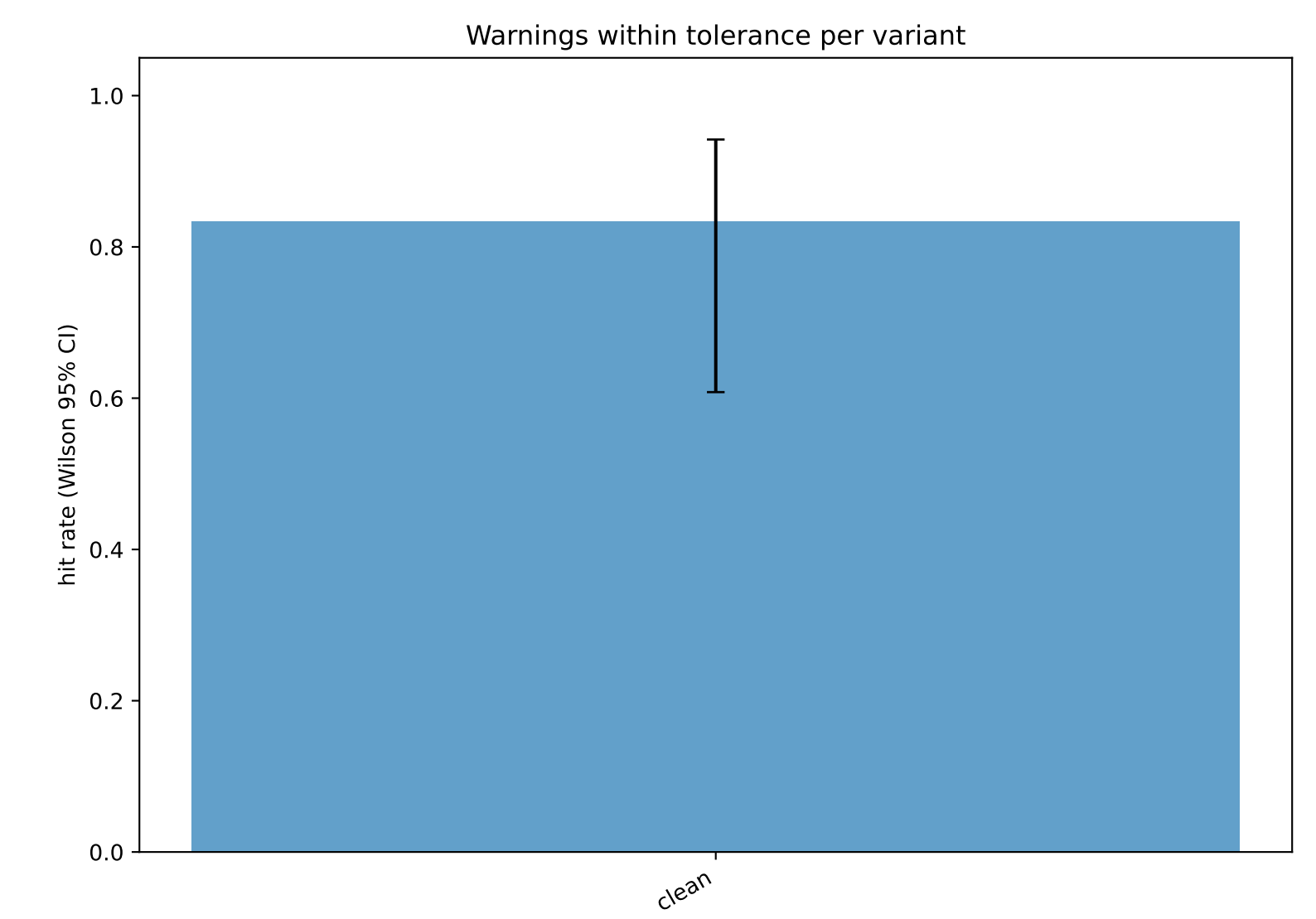
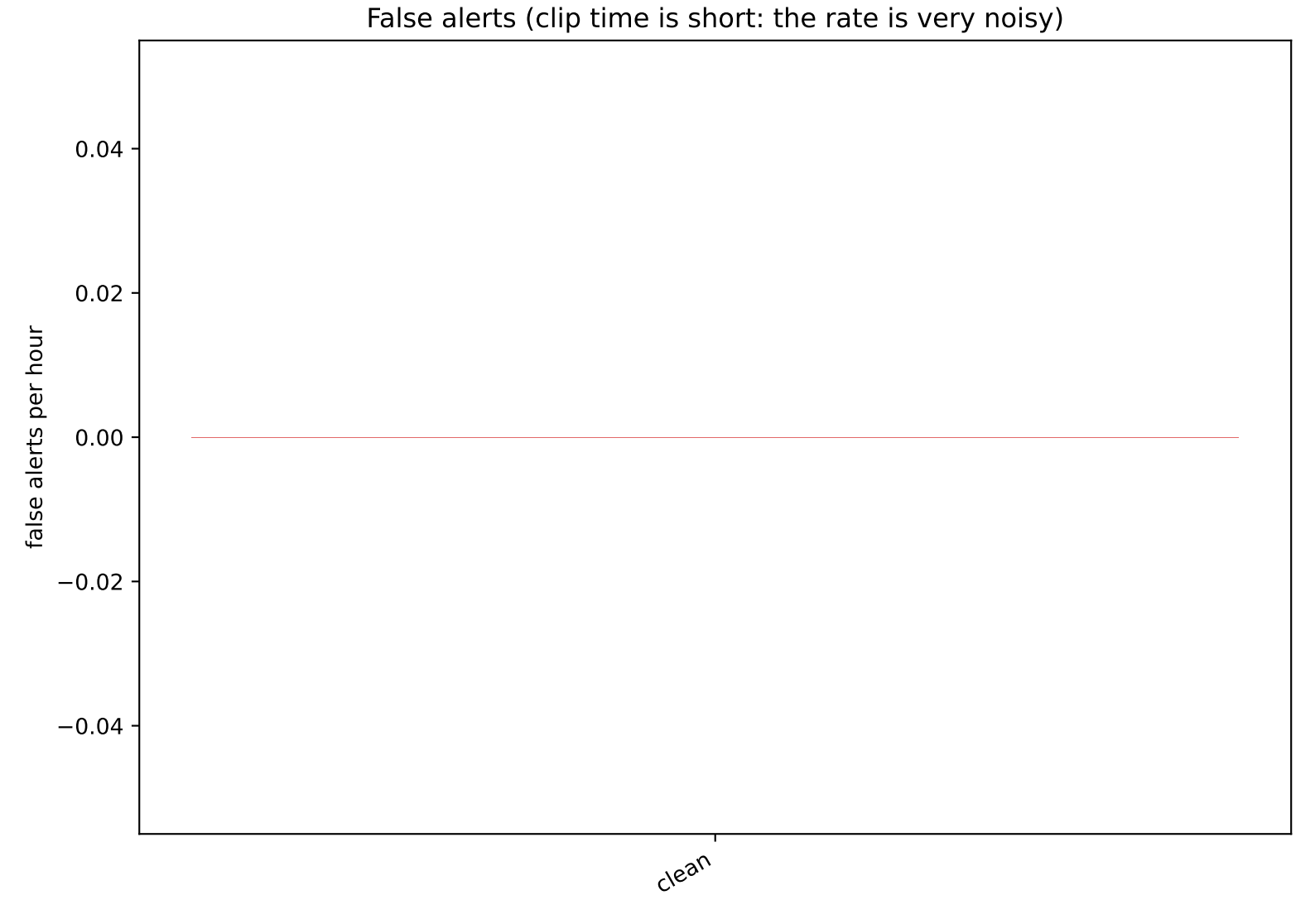
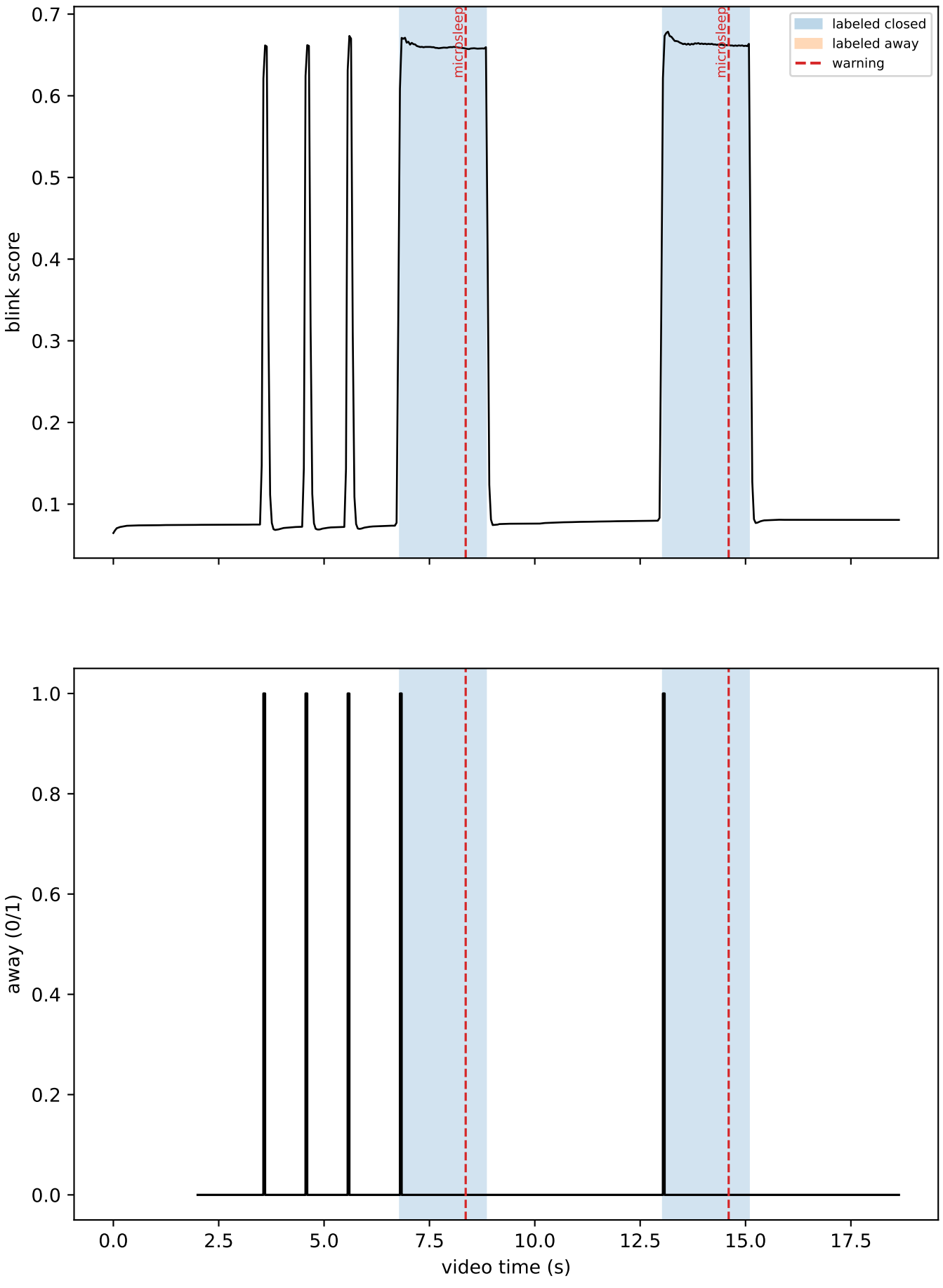
In-cabin DMS test bench: SD-202-style warning timing

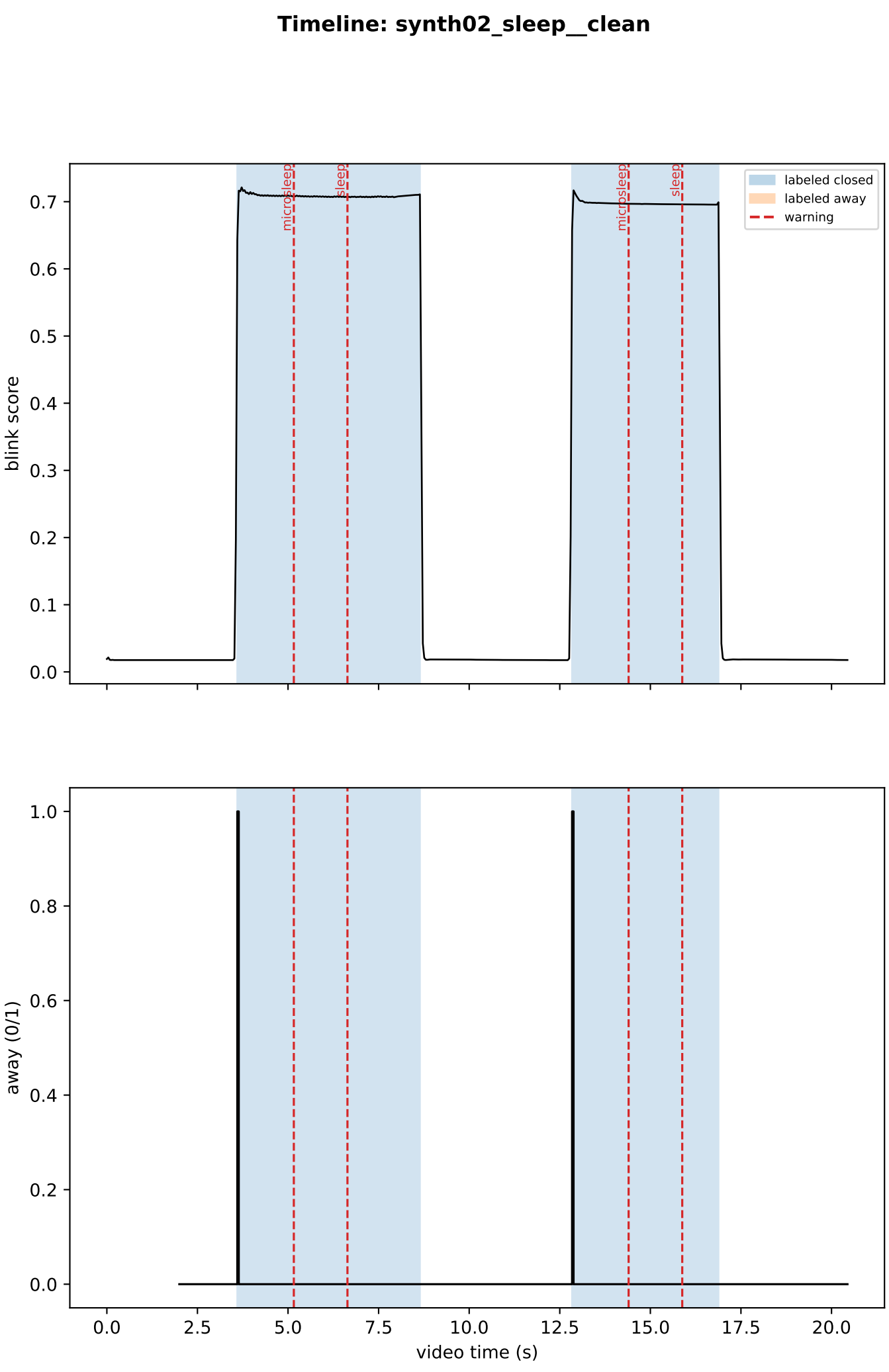
Method: MediaPipe Face Landmarker blendshapes (eye blink, head pose) with per-driver calibration give per-frame 'eyes closed' and 'gaze off road'. SD-202 timers (microsleep 1.5 s, sleep 3.0 s, long distraction 3.0 s) raise warnings. Each warning is compared with hand-labeled events; a warning is a hit if it is within 0.2 s of the required time. Intervals are Wilson 95%. False alerts are warnings that match no expected warning. Eyes are 'unknown' (not counted as closed) when the eye region is much darker than the forehead (dark lenses) or MediaPipe lost the face while YuNet still sees it. Arms (_ear = EAR instead of the blink blendshape, _mobilegaze = MobileGaze instead of head pose + eyeLook) are separate rows and compared on the arm comparison page.

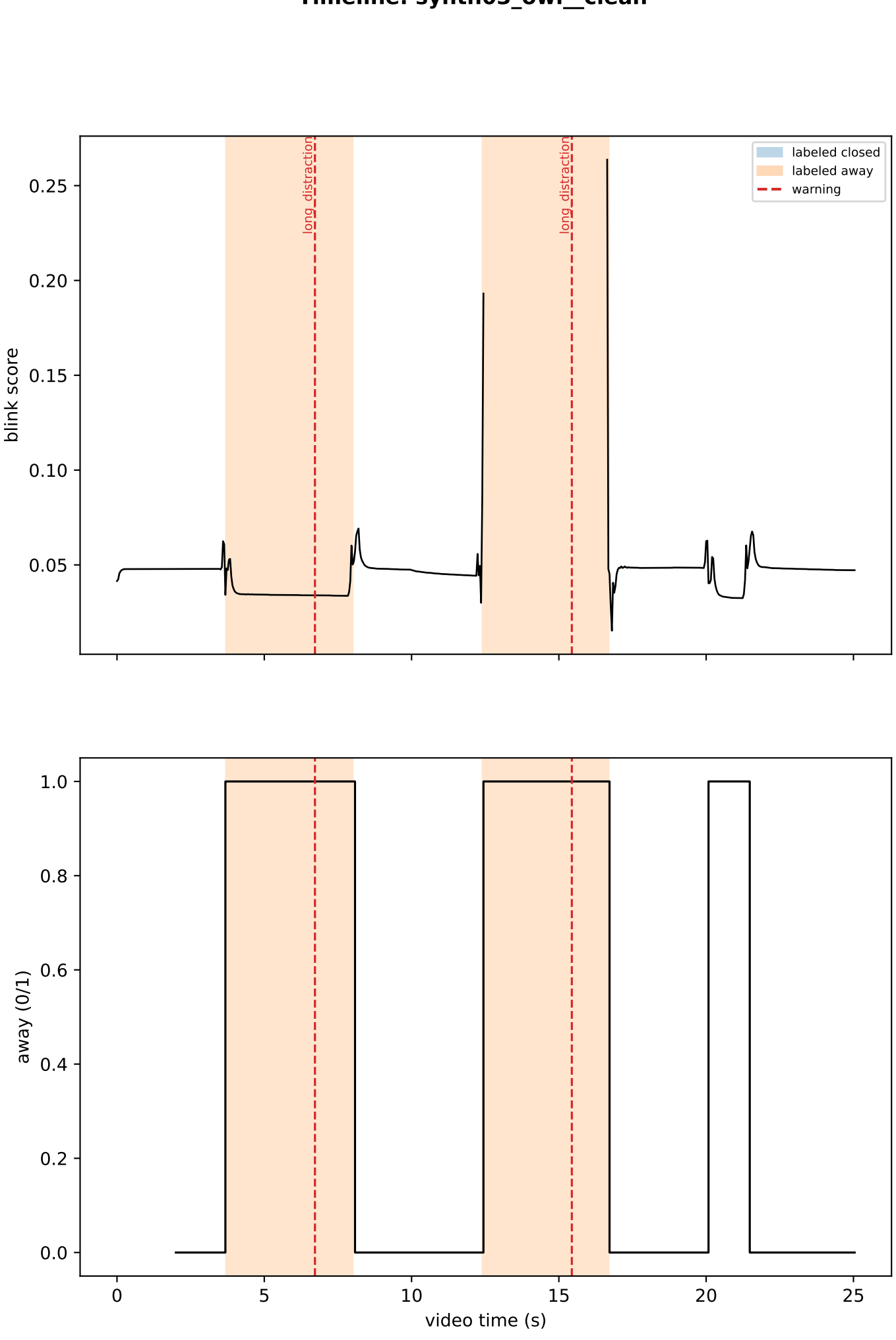
variant	hits/expected	hit rate [95% CI]	median |err| s	false alerts	false alerts/h	eyes unknown
clean	15/18	0.83 [0.61, 0.94]	0.04	0	0.0	0.23

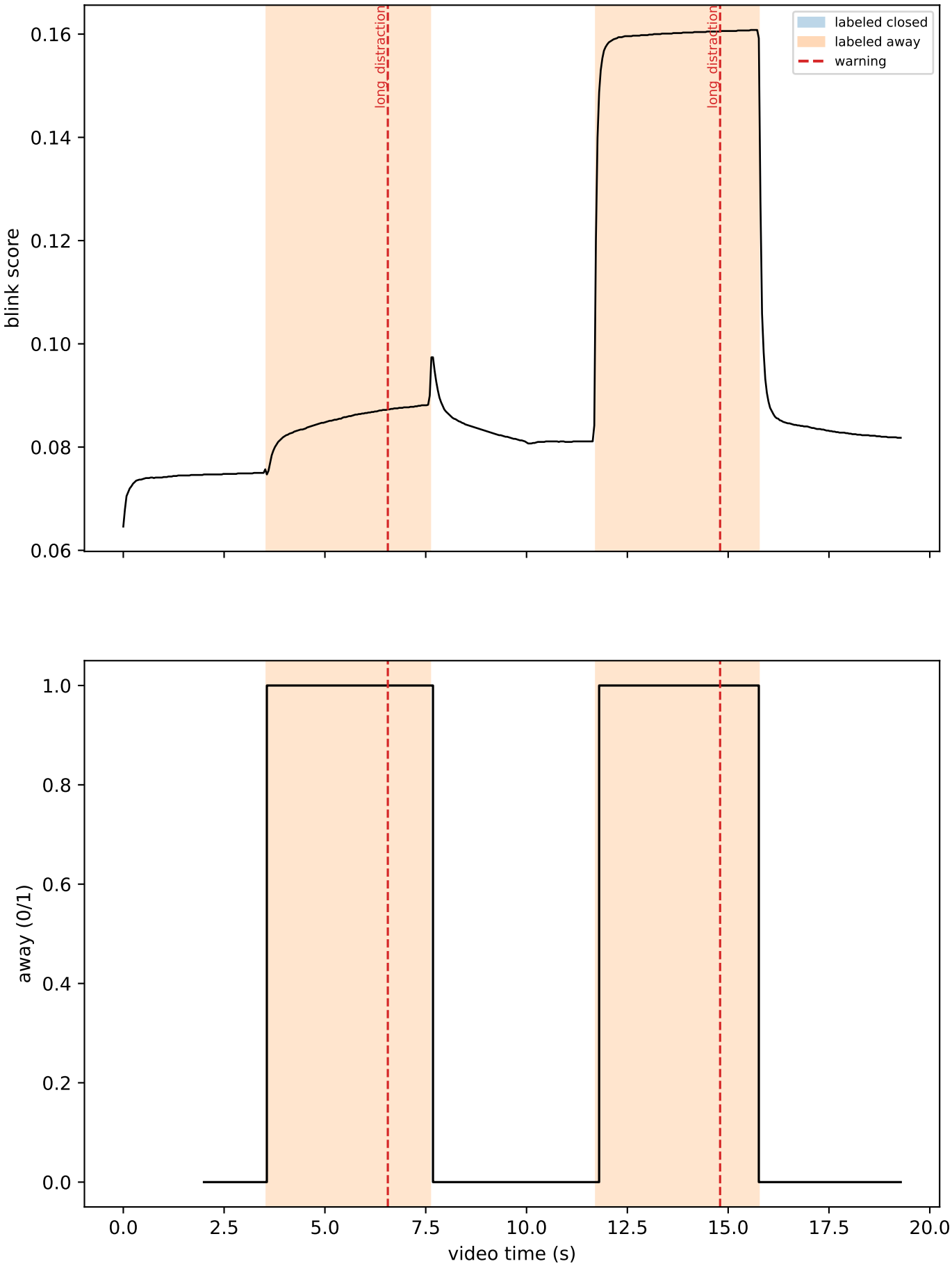

Warnings within tolerance per variant
hit rate (Wilson 95% CI)
1.0
0.8
0.6
0.4
0.2
0.0
clean

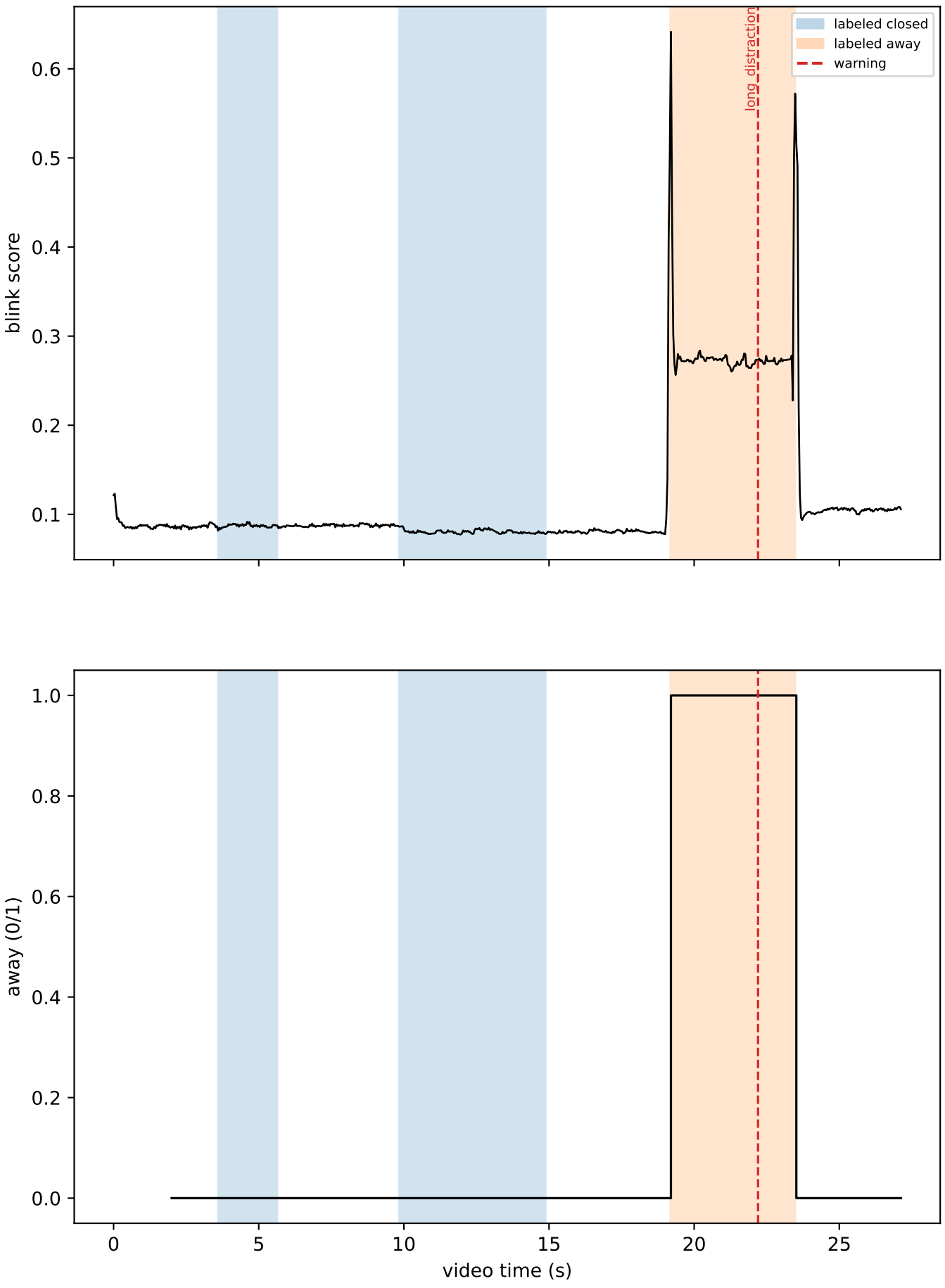

False alerts (clip time is short: the rate is very noisy)
false alerts per hour
0.04
0.02
0.00
−0.02
−0.04
clean

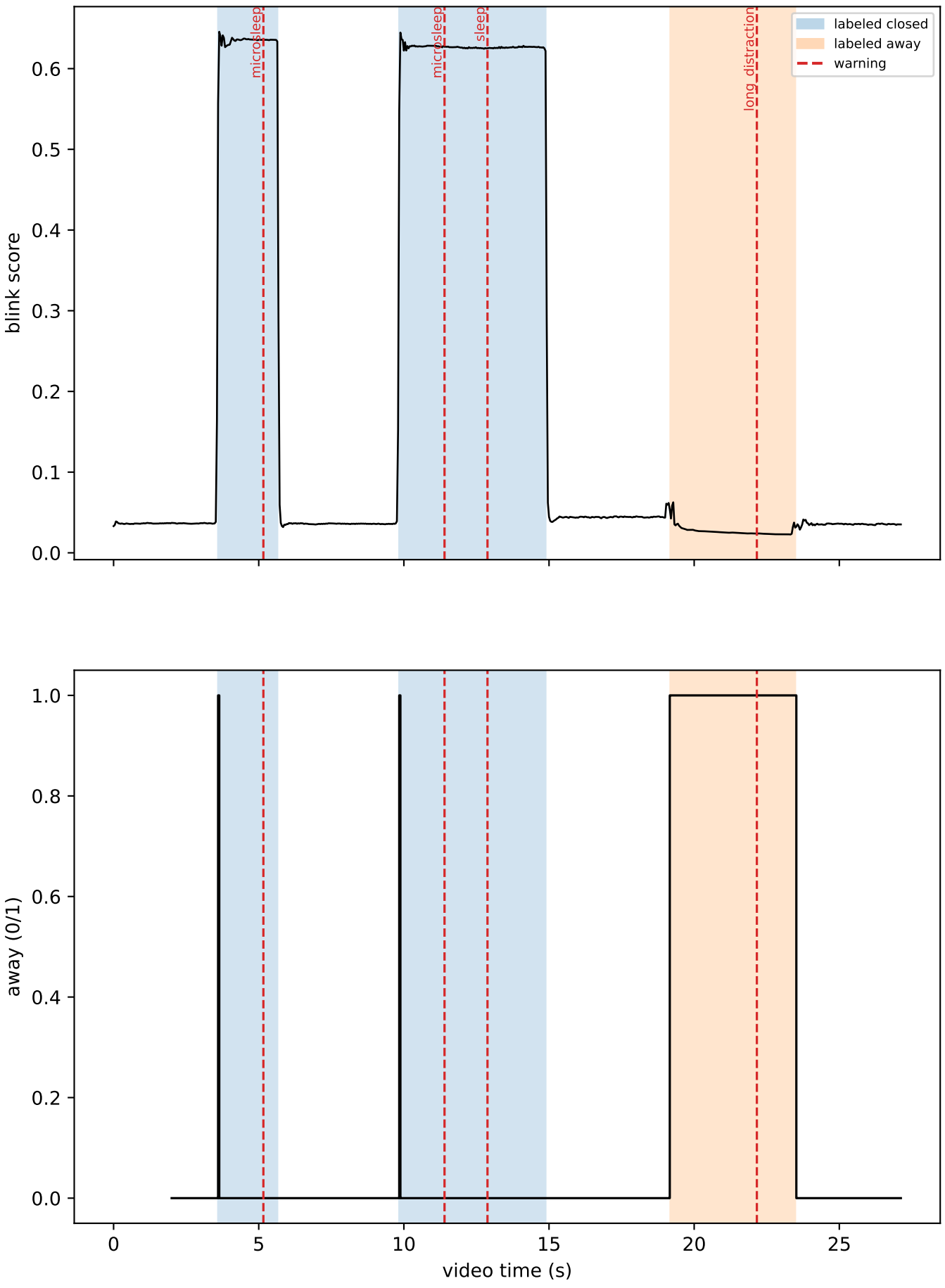

Timeline: synth01_microsleep__clean
blink score
0.7
0.6
0.5
0.4
0.3
0.2
0.1
microsleep
microsleep
labeled closed
labeled away
warning
away (0/1)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
video time (s)


Timeline: synth02_sleep__clean
labeled closed
labeled away
warning
microsleep
sleep
microsleep
sleep
blink score
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
away (0/1)
0.0
0.2
0.4
0.6
0.8
1.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
video time (s)


Timeline: synth03_owl__clean
labeled closed
labeled away
warning
long_distraction
long_distraction
blink score
0.25
0.20
0.15
0.10
0.05
away (0/1)
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25
video time (s)


Timeline: synth04_lizard__clean
labeled closed
labeled away
warning
long_distraction
long_distraction
blink score
0.16
0.14
0.12
0.10
0.08
0.06
away (0/1)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
video time (s)


Timeline: synth05_sunglasses__clean
labeled closed
labeled away
warning
long_distraction
blink score
0.6
0.5
0.4
0.3
0.2
0.1
away (0/1)
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25
video time (s)


Timeline: synth06_mask__clean
labeled closed
labeled away
warning
microsleep
microsleep
sleep
long_distraction
blink score
away (0/1)
video time (s)

Feasibility and limitations

Self-recorded tiny dataset: one subject, a phone camera, RGB (not IR). The number of events per variant is small, so the Wilson intervals are wide; a hit rate here is a functional check of the pipeline, not an estimate for other people or cars.

The variants (ir, glare, dark, sunglasses, mask, fps25) are image effects applied to the same clips, not newly rendered or newly recorded people. Real IR sunglasses, real mask fit and real night cabins will differ (domain gap).

Labels are made by hand by stepping through video frames, so label error is about one frame (0.04 s at 25 Hz). The tolerance of 0.2 s is therefore only slightly larger than the label and frame uncertainty.

Thresholds (microsleep 1.5 s, sleep 3.0 s, long distraction 3.0 s) are the values chosen in this project inside the SD-202 ranges; they are not those of any product. Euro NCAP scoring, other scenarios (owl/lizard glances, unresponsive driver) and vehicle intervention are not covered.

This is a test bench built for learning and demonstration. It is not a validation of a product and makes no product claim.